

Historic Building Information Modelling (HBIM)

What this tool produces · *A parametric, 3D digital model of a historic building, semantically enriched with historical documentation, material diagnostics, and long-term maintenance schedules.*

G4 · Proof of Sustainability

Dimensions

Heritage

Spatial

Economic

Preparing the workshop · *Historic Building Information Modelling (HBIM)*

G4 · Proof of Sustainability

Why this tool matters

Transforming static, decaying stones into a living digital memory, where every crack, material, and historical intervention is documented and protected against the erasure of time.

Evidence we produce

A parametric, 3D digital model of a historic building, semantically enriched with historical documentation, material diagnostics, and long-term maintenance schedules.

Duration & Material

Duration · Several months for scanning, processing, and semantic modeling

Material · 3D Laser scanners, UAVs, HBIM software (e.g., Revit, Archicad), Common Data Environment platforms

Reference case

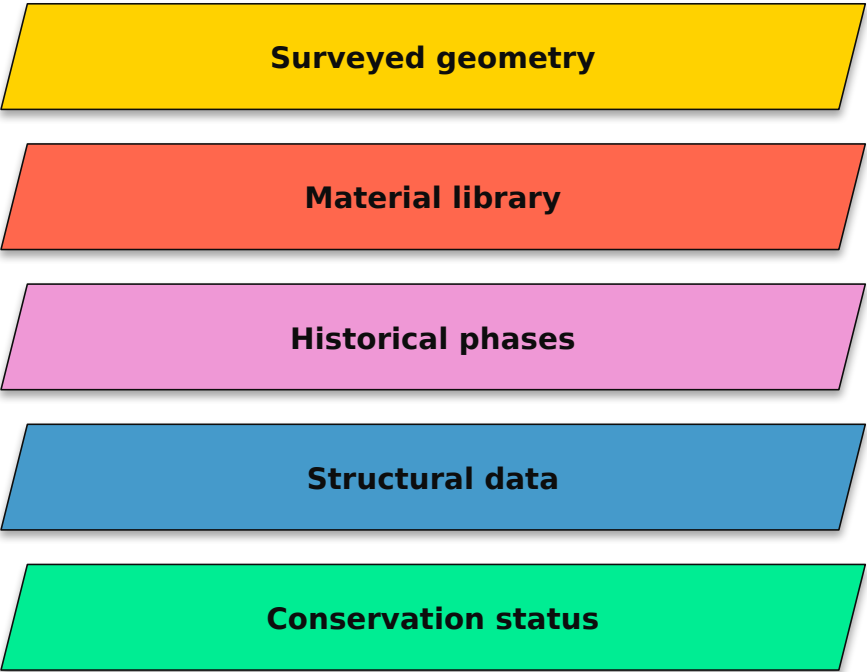
The restoration of the former Church of San Quirico all'Olivo in Lucca (Italy) utilized HBIM methodologies. Terrestrial laser scanning provided the geometric base, while the resulting HBIM model centralized heterogeneous historical and structural data, supporting the interdisciplinary design and long-term management of the restoration process.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Layered 3D model · heritage informed

Duration
Several months for scanning, processing, and semantic m...

Information layers



Data sources × layers

Laser scan	☐	☐	☐	☐	☐
Drone photogrammetry	☐	☐	☐	☐	☐
Archive drawings	☐	☐	☐	☐	☐
Material samples	☐	☐	☐	☐	☐
Oral history	☐	☐	☐	☐	☐

Status: collected · verified · published · referenced · LOD200+

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.